Smart Rover

Project Report

By

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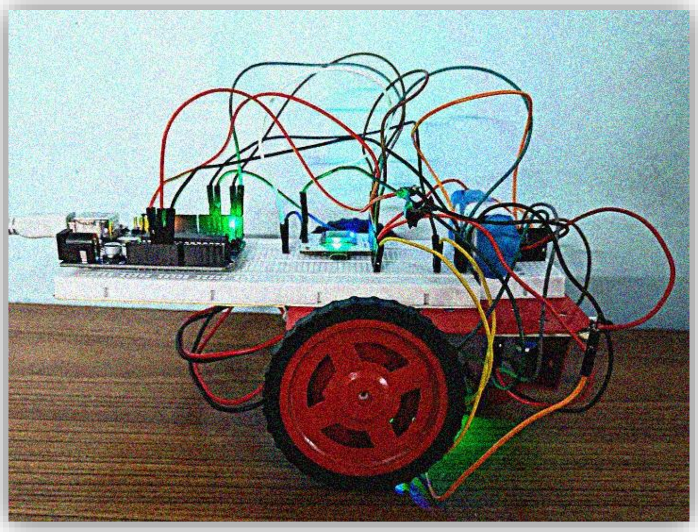


Background:

The SmartRover is a cloud-controlled vehicle with obstacle detection capabilities. Its inspiration comes from the Advanced Driving Assistant Systems (ADAS) which has one of the functions being automatic brake on close proximity to obstacles. This Project is just a simple integration of two concepts, Internet Controlled Vehicle and ADAS.

Aim:

To make an obstacle detection vehicle which is remote controlled through Cloud Dashboard using IR sensors.

Components:

Hardware Components:

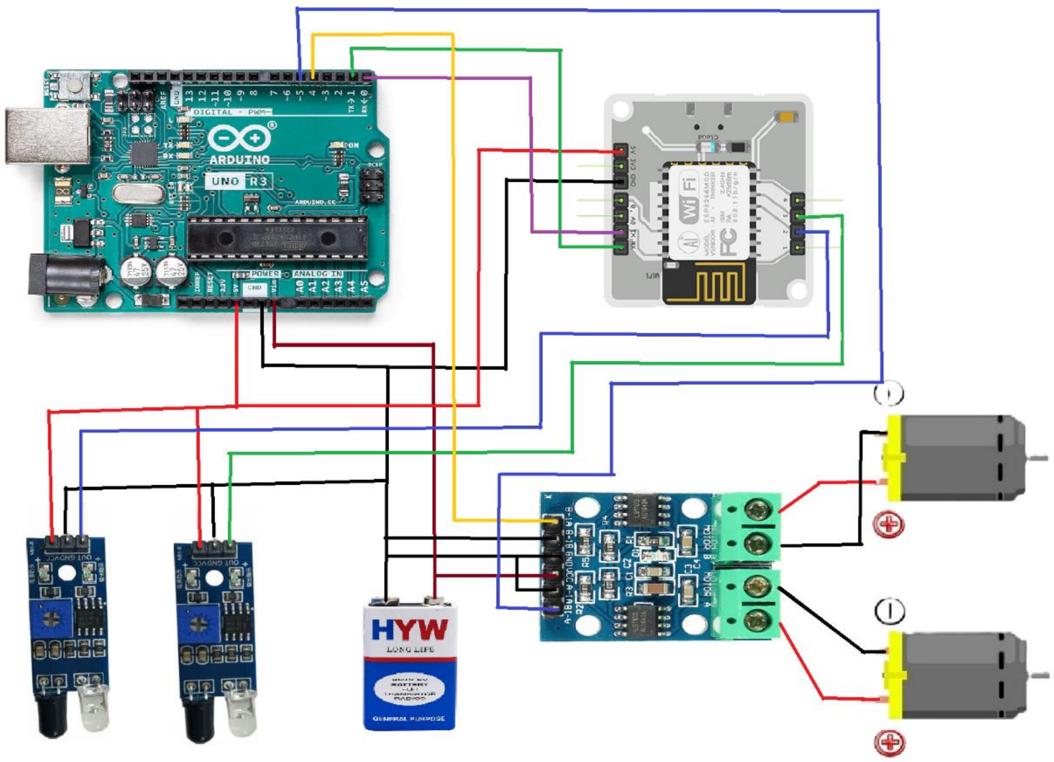
1. Arduino Uno
2. Boltiot Wifi Module
3. DC Motors
4. Motor Driver
5. Infrared Sensors
6. 9V Battery

7. Jumper wires (M-F & M-M)
8. Chassis
9. Universal Ball Roller
10. Wheels
11. Breadboards

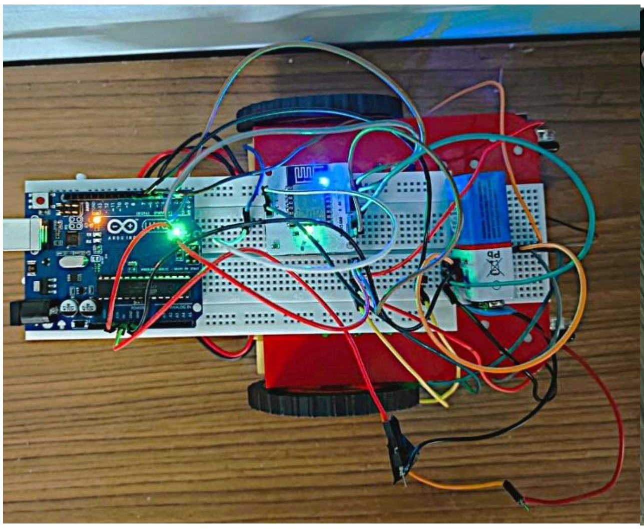
Software:

1. Arduino IDE
2. BoltIot Cloud

Circuit Connection:



1. Connect the two DC Motors to the Motor Driver. Connect VCC to 9V battery supply, GND and negative terminal of motors to ground and positive terminals of motors to pin 4 and 5 of Arduino.
2. Connect VCC of the two IR sensors to 5V supply of Arduino, GND of the two IR sensors to GND of Arduino and Outputs of IR sensors to pin 2 and 3 of BoltIot Wi-Fi Module.
3. Connect Tx of BoltIot Wi-Fi Module to Rx(pin0) of Arduino, Rx of BoltIot Wi-Fi Module to Tx(pin1) of Arduino, 5V to 5V supply of Arduino and GND to GND of Arduino.
4. Connect the 9V Battery to Vin of Arduino.
5. Mount all components on the Chassis including wheels, breadboard and roller.



Software Setup and Code:

GitHub Repository for webpage and Arduino: [ProjectGitHubRepository]

Bolt Setup:

- Open Bolt Cloud and add a product. Choose input devices and GPIO in the options below.
- Link Bolt device to the product.
- In Configurations Hardware tab, add two input variables for each IR sensor namely irvalue1 and irvalue2.
- In the code tab we will enter the html code from given GitHub repository.
- Save and deploy configuration.

Arduino Setup:

- Enter the Arduino Code from the given GitHub repository.
- Build code and upload to Arduino.

Observations and Output:

When we open the bolt app, we see a user interface with directions, a stop button and sensor readings. On tapping the buttons, the bot moves in the respective direction. On clicking stop, the bot stops immediately. Since there is no obstacle, no warning is given.

Now if we introduce an obstacle such as hand, the bot stops immediately.

If both IR sensors are triggered, the app says “Caution Obstacle Ahead”.

If right IR sensor is triggered, the app says “Obstacle on the right Go Left”.

If left IR sensor is triggered, the app says “Obstacle on the left, Go Right”

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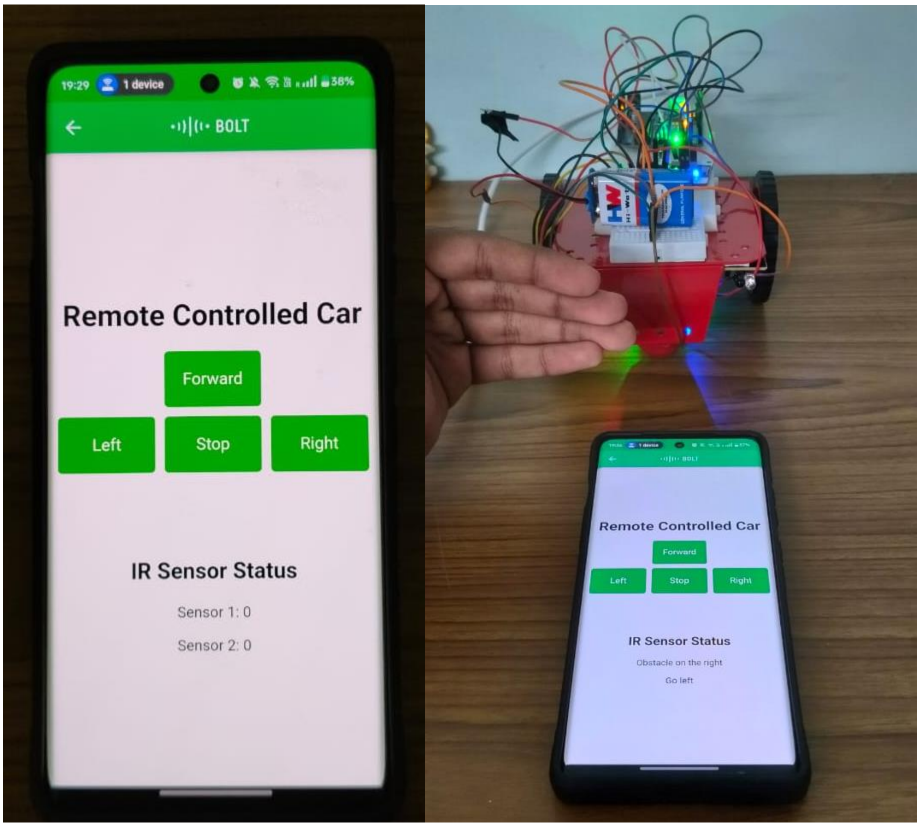

Conclusion:

This project serves as a foundational step towards more advanced autonomous systems, and the knowledge gained here can be applied to future endeavours in robotics, automation, and IoT-based applications. Through this project we see how to interface a rover/bot with the cloud. This forms a basic Cyber Physical System. Using more sophisticated equipment and more refined designs, we can develop ADAS systems which can be controlled remotely. The possibilities with this design development are limitless. By adding AI, Computer Vision and Machine Learning Algorithms we can develop autonomous rovers, vehicles and drones. The Obstacle Detection Aspect of the rover displays how a vehicle can traverse through highly dynamic environments on its own discretion.

Hence, the objective(aim) of this project is successfully demonstrated.